

Evaluating RAG Systems

Why evaluation is hard

Evaluating a retrieval-augmented generation system is harder than evaluating a classifier because the output is open-ended text rather than a label. A good evaluation separates the retrieval stage from the generation stage, because a wrong answer can come either from retrieving the wrong context or from the model misusing correct context. Measuring them together hides which stage is at fault.

Retrieval metrics

Context precision measures the fraction of retrieved chunks that are actually relevant to the question. Context recall measures the fraction of all relevant chunks that were successfully retrieved. Precision falls when irrelevant chunks are returned; recall falls when relevant chunks are missed. Tuning the number of retrieved chunks, the chunk size, and the retrieval strategy trades these off.

Generation metrics

Faithfulness measures whether the generated answer is supported by the retrieved context, that is, whether it avoids hallucination. Answer relevance measures whether the answer actually addresses the user's question rather than drifting to a related topic. A faithful but irrelevant answer and a relevant but unfaithful answer are both failures, so both metrics matter.

LLM-as-judge

Many modern evaluation frameworks, such as Ragas and DeepEval, use a strong language model as an automated judge to score faithfulness and relevance. This scales far better than manual review, though the judge model's own biases must be kept in mind. The recommended workflow is to curate a small set of representative question and answer pairs, run the system against them, and track the metrics over time so that regressions are caught before users notice them.

A practical checklist

First, build a golden set of real questions with known good answers. Second, log every retrieval and every generation so failures can be traced. Third, measure retrieval and generation separately. Fourth, re-run the evaluation on every meaningful change. Fifth, watch for latency and cost alongside quality, since a correct system that is too slow or too expensive is not usable in production.